`impop`$_k$ hands-on tutorial

IBS, IBD, and local ancestry from pangenome alignments

MEMPANG workshop

Time. Around 40-45 minutes. Compute time across the entire tutorial is a few seconds; the rest of the time is reading outputs, plotting, and answering interpretation questions. Work alone or in pairs (in pairs is good). Once finished, we will do a wrap-up altogether.

Where the data came from. The similarity matrices are obtained from `impg similarity` on PAF alignments. These are identities per window (in this case, fixed size, 10kb). The two TSVs in `data/` are exactly that, precomputed on `chr12`, so you don't pay the cost again today.

Why chr12 long arm. Every section below works on the long (q) arm of `chr12`, roughly 40 to 130 Mb. The arm gives us enough length to resolve recombination breakpoints visually.

Learning objectives

- read a per-window pairwise identity matrix from `impg similarity` and understand its structure
- detect identity-by-descent (IBD) segments between haplotypes using a 2-state HMM
- infer local ancestry of admixed query haplotypes against an HPRCv2 reference panel using a K-state HMM
- paint each grandchild's haplotypes against the four grandparents and resolve which homolog is paternal vs. maternal
- distinguish Viterbi decoding (most-likely path) from forward–backward posteriors (per-window marginals)

What is impop_k?

`impop_k` is part of the impop suite, a package for population genomics from implicit pangenomes. `impop_k` focuses on relatedness and ancestry inference (IBS, IBD, ancestry, jacquard) and operates directly on per-window pairwise identity matrices, the kind produced by `impg similarity` on a pangenome. It does relatedness, IBD, local ancestry, and kinship inference without ever calling variants, phasing, or building a VCF. Each binary wraps a hidden Markov model whose emissions are conditional probabilities over per-window identity, with transitions calibrated by recombination rate. The 2-state IBD detector marks segments as IBD vs. not-IBD; the K-state local-ancestry binary assigns each window to one of K reference populations. Swap the panel file and the same `ancestry` HMM paints chromosomes against any K labelled groups, including the four grandparents of a pedigree.

1. Setup

Set up the working directory and clone the repo:

```
$ ssh group@vesuvio.genenetwork.org
$ cd $DIR_MEMPANG26
$ git clone https://github.com/MarsicoFL/memimpopk
$ cd memimpopk
```

Library path

The shipped `bin/{ibs,ibd,ancestry,jacquard}` binaries are Linux x86_64 ELF and need `libgcc_s.so.1` at run time. Inside the `mempang-workshop` Guix environment that library is propagated by `(list gcc "lib")`, but `guix shell` does not auto-export `LD_LIBRARY_PATH`. Set it once before invoking the binaries:

```
$ export LD_LIBRARY_PATH="$GUIX_ENVIRONMENT/lib${LD_LIBRARY_PATH:+:$LD_-
LIBRARY_PATH}"
```

Smoke-test the environment:

```
$ bash test_setup.sh
```

A successful run prints `OK` and exits 0. The script verifies that the pre-built binaries execute on the host, that `python3` (≥ 3.8) is available, and that all data files are in place.

All commands below *must* be run from the `memimpopk/` directory.

2. Part 1 — IBS: looking at identity windows

The dataset

`data/identity_chr12_pedigree.tsv` contains pairwise identity per 10-kb window for the CEPH 1463 platinum pedigree on `chr12:40-130 Mb` (the long q arm). Eighteen haplotypes total: five full siblings (NA12879, NA12881, NA12882, NA12885, NA12886), the two paternal grandparents (NA12889, NA12890), and the two maternal grandparents (NA12891, NA12892). The two grandparental couples are mutually unrelated.

```
$ head -3 data/identity_chr12_pedigree.tsv
$ wc -l data/identity_chr12_pedigree.tsv
```

Columns: `chrom, start, end, group.a, group.b, estimated.identity, group.a.length, group.b.length`. The `group` columns are haplotype identifiers in `<sample>#<hap>` format. About 1.4M pair-window rows.

As a reference, you can see the pedigree structure below:

Run the explorer

```
$ python3 code/01_explore_ibs.py
```

This loads the TSV, splits the rows into intra-individual pairs (e.g. `NA12879#1` vs. `NA12879#2`) and inter-individual pairs (everything else), and writes `figures/ibs_histogram.svg`. To inspect it, copy the file to your local machine with `scp`:

```
$ scp group@vesuvio.genenetwork.org:/home/group/memimpopk/figures/ibs_histogram.svg
.
```

Alternatively, you can browse the pre-rendered figures in the repo at https://github.com/MarsicoFL/memimpopk/tree/main/tutorial.

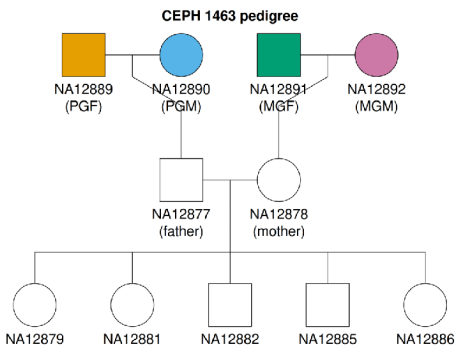

Figure 1: Platinum pedigree, three generations. Circles denote females, squares males. The four founders (grandparents) are colour-coded; each descendant haplotype is therefore a mosaic of founder chromosomes.

1.1

The right tail of the histogram concentrates at identity ≥ 0.9995. What biological scenarios produce values that high? Name two distinct ones.

1.2

In this pedigree-only dataset, the intra-individual and inter-individual means are extremely close, and the inter-individual mean is in fact *slightly higher*. That is counter-intuitive. Why does it make sense for this cohort?

1.3

A 10 kb window with `estimated.identity` ≈ 0.12 means $\sim$88% of bases are unaligned or mismatched. Discuss with your partner (self-partnership also works) how plausible each is as a *sole* cause, and justify: (a) multiple independent SNPs, (b) a > 5 kb deletion in one haplotype.

3. Part 2 — IBD detection

The IBD detector is a 2-state hidden Markov model: each window is in state `IBD` or `not-IBD`, transitions between adjacent windows are calibrated by recombination rate, and the emission $P(\text{identity} \mid \text{state})$ is learned by Baum–Welch from the data. The flags below (`-p-enter-ibd`, `-baum-welch-iters`) directly control these two HMM ingredients. See the talk slide that diagrams the model for the picture.

What we expect to find

Full siblings share, on average, 50% of the genome IBD. On a single 90 Mb arm we therefore expect several multi-Mb IBD segments per sibling pair. Between **each sibling and their paternal grandparents** (NA12889/NA12890) the expected mean fraction is 25%; same between each sibling and their maternal grandparents (NA12891/NA12892). Between the two grandparental couples, expected IBD is essentially zero.

Run the IBD detector

```
$ bash code/02_run_ibd.sh
```

This invokes `bin/ibd` on the pedigree identity matrix and writes `solutions/ibd_segments.tsv`. The relevant flags:

```
-similarity-file data/identity_chr12_pedigree.tsv
-region chr12:40000001-130000000
-size 10000
-min-len-bp 2000000
-min-lod 3.0
-baum-welch-iters 20
```

- `-min-len-bp 2000000` sets the minimum reported segment length to 2 Mb. This threshold ties directly to a key concept: the expected length of an IBD segment depends on the number of generations back to the most recent common ancestor (MRCA), as $E[L] \approx 1/(2g)$ Morgans. A 2 Mb segment ($\sim$2 cM) is therefore consistent with an MRCA roughly 25 generations ago — around 700 years.
- `-min-lod 3.0`: minimum LOD score $\log_{10}[P(\text{data} \mid \text{IBD}) \,/\, P(\text{data} \mid \neg\text{IBD})]$. A value of 3 means the segment is $10^3\times$ more likely under IBD.
- `-baum-welch-iters 20`: EM iterations for emission parameter learning.

Read the output

```
$ head -20 solutions/ibd_segments.tsv
```

Columns: `chrom, start, end, group.a, group.b, n_windows, mean_identity, mean_posterior, ..., lod_score`. Each row is one detected IBD segment between one haplotype pair.

2.1

Pick a sibling pair (any two of NA12879, NA12881, NA12882, NA12885, NA12886) with ≥ 2 multi-Mb IBD segments on this arm. Report the intervals and LOD scores, and state the *minimum* number of meiotic crossovers in the parental lineages that the segment boundaries imply. Justify your count.

2.2

There are detected "IBD" segments between `NA12892#1` and `NA12892#2`, the two haplotypes of one person, and also between the two haplotypes of *other* grandparents and even of some siblings. Strict IBD between the two haplotypes of one diploid individual would imply autozygosity. What conservative explanation should you consider first, and what would you check to discriminate?

2.3

For each of the two HMM transition knobs, predict the direction of change in (i) segment count, (ii) mean segment length, (iii) total IBD Mb covered, when you raise `-p-enter-ibd` from 10^{-4} to 10^{-2}, and separately when you drop `-expected-seg-windows` from 50 to 5. Which knob has the largest effect on *total* IBD Mb, and why?

4. Part 3 — Local ancestry inference

How the chimeras and the panel were built

Five chimeric query haplotypes (`CHIM_00#1` ... `CHIM_04#1`) were stitched from real HPRCv2 assemblies along ground-truth tracts: each combines one AFR donor (HG01884) with one of five EUR donors (HG00097, HG00099, HG00126, HG00128, HG00133). The reference panel in `data/populations.tsv` is a different set of 20 HPRCv2 haplotypes (10 AFR + 10 EUR). **None of the panel samples were used as donors of the chimeras.**

The dataset

`data/identity_chr12_admix.tsv` contains pairwise identity per 10-kb window for `chr12:50-120 Mb` — 70 Mb in the middle of the long q arm. Because 70 Mb is much longer than a typical admixture tract, each chimera carries 3–7 ground-truth breakpoints in this window.

Run local ancestry inference

```
$ bash code/03_run_ancestry.sh
```

The relevant flags:

```
-similarity-file data/identity_chr12_admix.tsv
-query-samples data/queries.txt
-populations data/populations.tsv
-region chr12:50000001-120000000
-window-size 10000
-auto-configure
-identity-floor 0.9
-posteriors-output solutions/ancestry_posteriors.tsv
```

- `-auto-configure`: estimates two emission-model knobs from the data, instead of asking you to set them.
  - *Softmax temperature* T: rescales per-window similarity differences before turning them into emission probabilities via $\text{softmax}(\Delta/T)$. Small $T \to$ sharp commit to one population from tiny similarity gaps; large $T \to$ smooth, conservative posteriors.
  - *Pairwise-contrast weight*: how much weight the emission gives to the *difference* between max-similarity-to-population-A and max-similarity-to-B, relative to the absolute identity. High contrast weight emphasizes which panel is closer; low contrast weight makes the emission depend more on raw identity values.
- `-identity-floor 0.9`: drops windows whose maximum similarity across the panel is below 0.9 — usually alignment-gap artifacts.

Inspect the painting

```
$ head -20 solutions/ancestry_segments.tsv
$ python3 code/04_plot_painting.py
```

This produces `figures/ancestry_painting.svg` with two bars per chimera: "truth" on top, "impop" below. Open it.

3.1

Pick one chimera and count its ground-truth breakpoints in the 50–120 Mb interval. How many of them does the model recover within ∼1 Mb? Which ones does it miss or hallucinate?

3.2

Identify one genomic interval (≥ 500 kb) where ≥ 3 of the 5 chimeras show the same kind of error (mis-call or smeared boundary). Propose one testable property of the reference panel in that interval that would explain it (e.g. panel ancestry imbalance, low diversity, segmental duplication).

3.3

Look at the `mean_posterior` column of the TSV. Several wrong calls carry posteriors ≥ 0.9. How can the model be confidently wrong?

5. Part 4 — Open the black box: forward–backward vs. Viterbi

The TSV from Part 3 is the *Viterbi* decoding: the single most likely sequence of ancestry states. The HMM also computes a *marginal posterior* per window via forward–backward: the probability of each state at each window, integrating over all paths. Part 3 already wrote these per-window numbers to `solutions/ancestry_posteriors.tsv`.

```
$ head -3 solutions/ancestry_posteriors.tsv
$ python3 code/07_plot_posteriors.py
```

This produces two figures:

- `figures/posterior_trace.svg` for `CHIM_03#1`: ground truth on top, P(AFR) per window in the middle (the marginal posterior), Viterbi-decoded segments at the bottom. The dashed line at $P = 0.5$ marks where Viterbi flips.
- `figures/panel_similarity.svg`: the raw input the HMM emissions see. Per window, Δ = (max identity to AFR panel) − (max identity to EUR panel), smoothed over 250 kb.

Notice how small the per-window contrast is: median $|\Delta|$ is only $1\text{–}3 \times 10^{-4}$. The HMM has to chain dozens of windows to make a confident call from such a weak per-window signal.

4.1

Find a window where P(AFR) sits between 0.4 and 0.6: the model is genuinely uncertain there. What is Viterbi's call at that same window, and what does that tell you about the difference between the marginal posterior (forward–backward) and the most-likely-path (Viterbi)?

6. Part 5 — Grandparent painting of the CEPH pedigree

We swap roles. Instead of a population panel, we use the *four grandparents* as four "populations" (K=4), and paint each grandchild's two haplotypes against them. This is the same HMM, same binary, same identity matrix as Part 2 — only the panel file changes.

The setup

```
$ cat data/pedigree_populations.tsv
$ cat data/pedigree_queries.txt
```

Four "populations", each containing the two haplotypes of one grandparent:

```
GP_PAT NA12889#1, NA12889#2 (paternal grandfather)
GM_PAT NA12890#1, NA12890#2 (paternal grandmother)
GP_MAT NA12891#1, NA12891#2 (maternal grandfather)
GM_MAT NA12892#1, NA12892#2 (maternal grandmother)
```

The ten queries are the ten haplotypes of the five grandchildren (NA12879, NA12881, NA12882, NA12885, NA12886 — `#1` and `#2` each).

Run the painting

```
$ bash code/05_paint_pedigree.sh
$ python3 code/06_plot_pedigree_painting.py
```

This produces `figures/pedigree_painting.svg`: two stacked bars per grandchild, one per haplotype, coloured by inferred grandparent of origin.

What you should see

The opening slide of the talk ("a chromosome inherited through a pedigree") is now a concrete result on real data. In particular:

- Each grandchild's two haplotypes split cleanly: one is a mosaic of *only* GP_PAT + GM_PAT blocks (paternal homolog), the other of *only* GP_MAT + GM_MAT blocks (maternal homolog).
- Breakpoints within a homolog mark actual meiotic recombinations in the mother or father of the CEPH family.
- The input data never told us which haplotype index (`#1` or `#2`) was paternal. The painting does.

5.1

For one grandchild of your choice, tell me which of `#1` and `#2` is paternal and which is maternal. Justify from the figure, not from the FASTA headers.

5.2

Across the 5 grandchildren you have 5 paternal homologs and 5 maternal homologs in the chr12:40–130 Mb window. Tally the breakpoints *separately* on the paternal-side and maternal-side homologs. Given that chr12q is roughly 80 cM, what crossover rate per meiosis does each tally imply? Is the gap between the two totals consistent with what you expect biologically, or is the sample too small to tell?

5.3

Suppose on one grandchild you saw a haplotype that alternates `GP_PAT` and `GP_MAT` blocks within 40–130 Mb. Name two distinct causes that could produce this pattern — one biological, one technical — and describe the smallest additional check you could run on `solutions/pedigree_painting.tsv` to distinguish them.

The end

Write a 1-paragraph summary covering (a) one thing you found in the IBS histogram that was not obvious before, (b) one IBD segment you can confidently call between two named haplotypes,

(c) one ancestry call from Part 3 where the model is wrong with high confidence and what that tells you about calibration, and (d) which grandchild haplotype indices (`#1` vs `#2`) you concluded to be paternal vs maternal from Part 5.